A Coffee Filter Parachute

01 What slows the parachute?



A wide canopy meets lots of air as gravity pulls the parachute down.

Look closer. Find the wide canopy above the hanging paperclip and notice the open space it covers.

A little explanation. Gravity pulls the parachute down. As it moves, tiny moving pieces of air push back. That air resistance, also called drag, slows the fall; it does not turn gravity off.

Try this. An adult stays beside you while you attach equal strings or yarn and a paperclip or small toy to a coffee filter. Ask that adult to use the scissors. Drop from standing height, or let the adult hold it higher. Never climb on chairs, stair railings, windows, or balconies.

02 Why does the flat filter fall more slowly?



The flat, wide filter has more surface area facing the air.

Look closer. Compare the open, wide filter with the same kind of filter crumpled into a small shape.

A little explanation. The wide filter has more surface area facing the moving air. It catches more air and makes more drag, so it falls more slowly than the crumpled filter even though both filters weigh the same.

Try this. Drop one filter flat and the same filter crumpled. Count the fall time or use a timer, then open the filter and repeat.

03 What changes when the load is heavier?



A second paperclip adds weight, so gravity pulls down harder while the same canopy provides drag.

Look closer. Look for one paperclip under one canopy and two paperclips under the matching canopy.

A little explanation. With the same canopy, a second paperclip adds weight, so gravity pulls down harder while the canopy area stays the same. In this test, that parachute usually descends faster. Without air, the extra paperclip would not make it fall faster.

Try this. Keep the filter and strings the same. Compare one paperclip with two paperclips over several drops.

04 How can a tiny hole make the fall steadier?



A tiny top hole can let air through and reduce side-to-side wobble.

Look closer. Find the small centered hole and watch the canopy stay open with less side-to-side wobble.

A little explanation. A small hole at the canopy top lets some air flow through. That can keep air from spilling unevenly around the edges, so the parachute may wobble less.

Try this. Predict what a tiny centered hole will change. Test the uncut canopy first, then ask an adult to make the hole and measure several more drops.